



PRODUCT BROCHURE & TECHNICAL CATALOG

Houston Anodes

Protecting the Infrastructure That Powers the World

Designing, Manufacturing and Delivering High-Quality Aluminum and Zinc Sacrificial Anodes Since 1977. Headquartered in the Energy Corridor of Houston, Texas. 100% Made in the USA.

47+

YEARS

1,000+

PROJECTS (5 YRS)

35 yr

NAVY SERVICE LIFE

0

CLAIMS IN 47 YRS

725

TONS / MONTH

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Standard sizes shown throughout this catalog. Any weight, size, shape, and configuration required by the customer can be manufactured. Weight based on aluminum alloy; also available in zinc alloy. When ordering, please specify metal and chemical composition.

DRAFT

WHO WE ARE

Protecting Offshore, Subsea and Onshore Assets and Infrastructure Since 1977

Houston Anodes has been one of the leading manufacturers of high-quality sacrificial aluminum and zinc anodes in the world since 1977. Today, Houston Anodes protects billions of dollars' worth of equipment and infrastructure worldwide.

Every step of our manufacturing process — from raw material testing and alloy formulation to casting, quality inspection, and delivery — takes place at our facility in the Energy Corridor of Houston, Texas.

Equipped with a state-of-the-art laboratory accredited to the ISO 17025 standard and superior electromagnetic induction technology. Management System Certified ISO 9001:2015 by DNV.

47+

Years

1,000+

Projects

0

Claims

725

Tons/Mo

Close to Our Customers

- Houston Energy Corridor headquarters — 2–4 week lead times
- Serving Top 100 energy companies across five continents
- 80% of PEMEX anodes for 35+ years
- U.S. Navy anodes with 35–year proven service life
- Approved FMC Global Supplier
- 100% Made in the USA

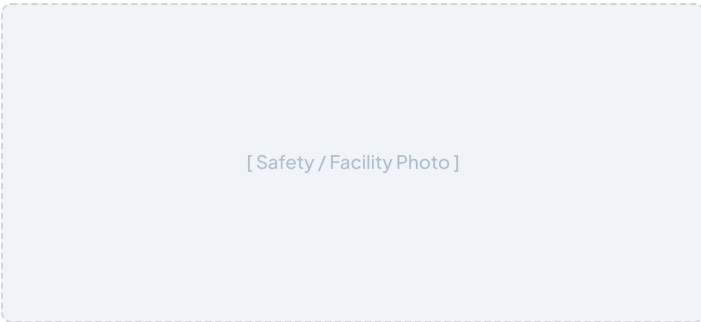


OUR #1 PRIORITY

Employee Safety & Development

At Houston Anodes International, we are proud of our holistic approach to safety in the workplace. To that end, we commit to ensure no harm comes to our employees, or damage to the environment.

We will endeavor to prevent all labor-related accidents and to continuously improve our HSE performance through the active participation and support of all our stakeholders. The successful management of HSE is an essential part of our business model and corporate culture.



Employee Development

Ongoing training for technical skills and safety awareness at every level.

Safer Technology

Induction eliminates open-flame and manual hand-stirring hazards.

Zero-Harm Culture

Prevention of all labor-related accidents as a core objective.

HSE Policy Highlights

- No open flames — electromagnetic induction generates heat without fire
- No manual stirring with steel rods — eddy currents homogenize the alloy automatically
- Eliminates risk of superheating metals
- Controlled temperature melting adjusted per metal type
- Continuous improvement through active participation of all stakeholders
- Prevention of all labor-related accidents and environmental damage

SAFETY IS OUR #1 PRIORITY

QUALITY IS OUR FOUNDATION

Quality Control & Laboratory

In-house state-of-the-art Laboratory equipped with a **Bruker-Quantron Magellan Q8 spectrometer**, the latest Photomultiplier for compelling analytical performance including analysis of non-metallic inclusions, configurable for virtually any metal analysis task.

- All raw materials inspected and tested upon arrival prior to release for production
- Chemical results verified against Mill Test Reports (MTRs)
- Chemical analysis performed on every heat
- Electrochemical properties tested on every batch according to NACE TM-0190
- Rigorous inspection and testing points throughout production
- Scheduled equipment calibration per OEM specifications
- Immediate, accurate test results



NOT A SINGLE CLAIM IN 47 YEARS

ISO 17025
Accredited in-house testing laboratory

ISO 9001:2015
Certified by DNV

NACE TM-0190
Electrochemical testing every batch

100% USA
All testing performed in-house

HIGHEST PURITY

Raw Materials

Houston Anodes fabricates anodes using only raw materials of the highest purity, sourced from premium U.S. and global suppliers. Every delivery is inspected and tested in our ISO 17025–accredited laboratory before release into production.

99.9%

Primary Aluminum

Purity

99.99%

Special High Grade Zinc

Purity

100%

Indium

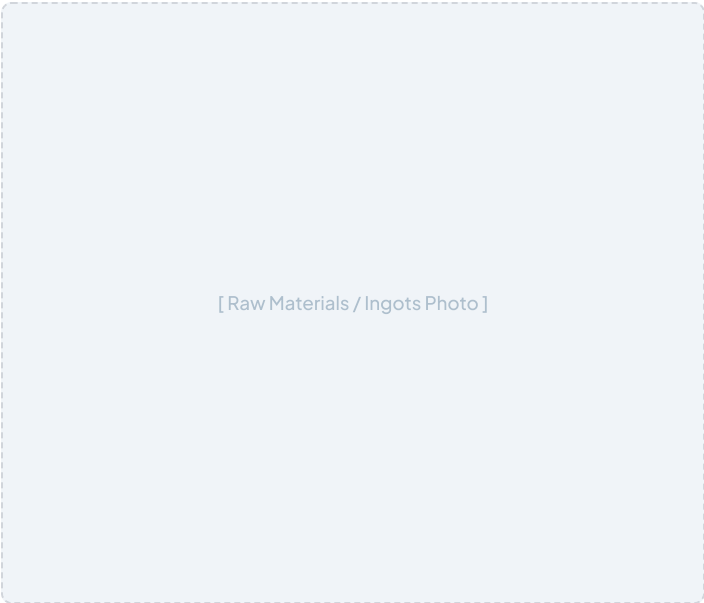
Purity

ASTM

Steel

Cores & structures

NEVER MELT SCRAP OR RECYCLED ANODIC MATERIAL



What Houston Anodes Delivers

- Right Chemical Composition

Proprietary aluminum alloys, high-purity zinc, custom formulations
- Right Electrochemical Properties

No porosity, perfect density. Compliant with NACE, DNV, ISO, NORSOK
- Right Weight

From 5 lbs to custom configurations
- Right Fit

Any application, any specification, manufactured to order. We make our own molds.

OUR MANUFACTURING ADVANTAGE

Electromagnetic Induction Technology

The furnace is composed of a deionized water-cooled copper coil. When energized with an AC power source, it develops an electromagnetic field. Any metal charge kept inside the crucible is subjected to electromagnetic induction.

The electromagnetic field creates eddy currents in the charge, causing an opposite circulating stirring flow of electrons within the molten mass. This generates heat in the charge depending on its resistivity — without flames and without touching the material.

Advantages

- 100% homogeneous alloy of exceptional electrochemical purity
- Controlled temperature adjusted to each metal's specific melting point
- Consistent batches and final product every time
- **Gas furnaces take 50% longer to melt and process**
- Non-contact melting — reliable, repeatable, extremely efficient
- Enhanced employee safety — no open flames, no manual stirring



725 Tons/Month Total	400kW Furnace 1: 300 t/mo	625kW Furnace 2: 425 t/mo
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Operating range: 0–2400V, 0–625kW, 0–2000Hz. Precise temperature control eliminates the risk of superheating the metals.

100% Made in the USA

PROVEN RESULTS

Induction vs. Gas Furnace Comparison

Gas Combustion Furnace

Inherited Deficiencies

- Direct flame transfers hydrogen & carbon — contaminating the alloy
- Hand stirring with steel rod — extremely dangerous — iron particles
- Heterogeneous mixture — indium only partially induced
- Passive/inert portions reduce anode lifespan
- 50% longer processing time

59%

Active material consumed

41%

Passive material wasted

Electromagnetic Induction (Houston Anodes)

Superior Performance

- Zero contamination — non-contact melting without flames
- 100% homogeneous alloy — complete indium activation
- Zero porosity, perfect density
- 40% longer anode lifespan
- 50% faster melting and processing

100%

Active material consumed

0%

Passive material wasted

[Gas Furnace Photo — for comparison]

[Induction Furnace Photo — Houston Anodes]

PRODUCT CATEGORY

Platform Anodes

Houston Anodes aluminum platform and structural galvanic anodes are cast in low-iron, high-purity aluminum alloys, and can be manufactured in a wide variety of configurations for new construction or retrofitting.

All standard configurations are shown in the following pages. Houston Anodes can cast anodes with custom trapezoidal cross-sections to any required specifications.

Configurations

- 90° Core — Type A & Type B
- 45° Core — Type A & Type B
- Straight Core — Type A & Type B

Weight Range

180 – 1,250 lbs (standard). Custom weights available.

Core Types

2" SCH. 80, 3" SCH. 80, 4" SCH. 80

ALUMINUM ALLOY

ZINC ALLOY

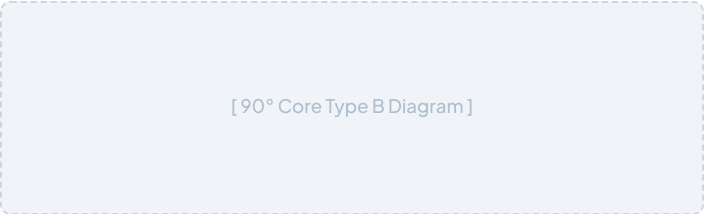
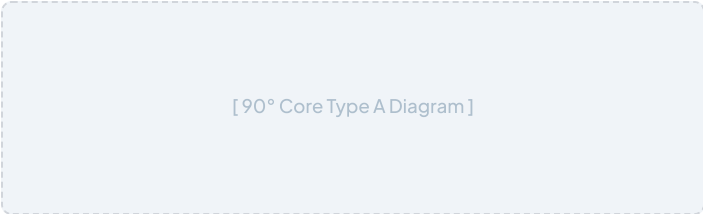
CUSTOM CROSS-SECTIONS



[Platform Anode Photo]



Platform 90° Core — Type A & Type B

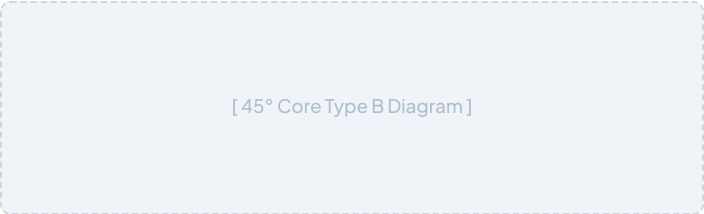
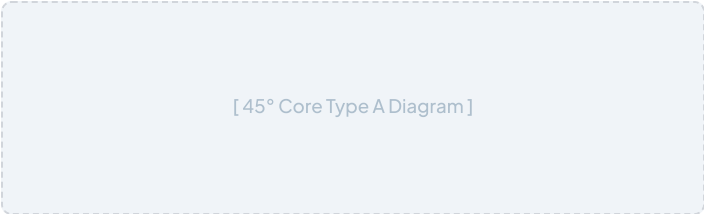


NAME	LA (IN)	LC (IN)	W (IN)	H (IN)	NET WT (LBS)	GROSS WT (LBS)	CORE
PF_180_AC90 / BC90	60	30 / 48	6½	5⅝	180	196–202	2" SCH.80
PF_285_AC90 / BC90	105–120	90–93	5½	5⅝–5¾	285	328–340	2" SCH.80
PF_325 (0–3)	60–120	30–100	5½–8¼	5¾–7¾	325	368–411	3" SCH.80
PF_407 / 411	72–96	42–84	7–8	7–8½	407–411	473–484	3" SCH.80
PF_540 (0–1)	96–98	66–86	6½–8	7¾–9¾	540	606–613	3" SCH.80
PF_725 (0–3)	95–98	65–86	9–10	9–10	725	813–881	4" SCH.80
PF_1050	144	114–120	9	9	1,050	1,187–1,200	4" SCH.80
PF_1100	122	92–98	10½	10	1,100	1,120–1,218	4" SCH.80
PF_1250	144	114–120	10	10½	1,250	1,387–1,400	4" SCH.80

Standard sizes shown. Any weight and size required by the customer can be manufactured. Weight based on aluminum alloy; also available in zinc alloy.



Platform 45° Core — Type A & Type B

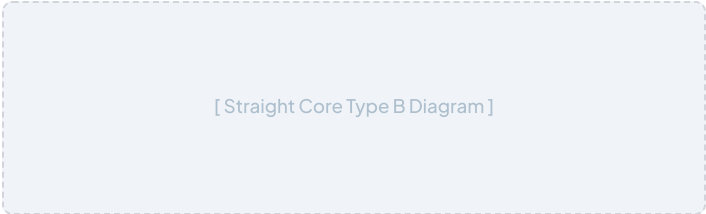
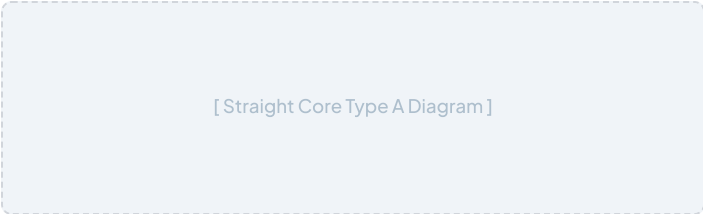


NAME	LA (IN)	LC (IN)	W (IN)	H (IN)	NET WT (LBS)	GROSS WT (LBS)	CORE
PF_180_AC45 / BC45	60	72-101	6½	5⅝	180	212-228	2" SCH.80
PF_285_AC45 / BC45	105-120	117-161	5½	5⅝-5¾	285	328-364	2" SCH.80
PF_325 (0-3)	60-120	72-161	5½-8¼	5¾-7¾	325	356-470	3" SCH.80
PF_407 / 411	72-96	84-137	7-8	7-8½	407-411	445-541	3" SCH.80
PF_540 (0-1)	96-98	108-139	6½-8	7¾-9¾	540	542-594	3" SCH.80
PF_725 (0-3)	95-98	107-139	9-10	9-10	725	791-913	4" SCH.80
PF_1050	144	156-192	9	9	1,050	1,245-1,290	4" SCH.80
PF_1100	122	134-163	10½	10	1,100	1,188-1,238	4" SCH.80
PF_1250	144	156-192	10	10½	1,250	1,445-1,490	4" SCH.80

Standard sizes shown. Any weight and size required by the customer can be manufactured.

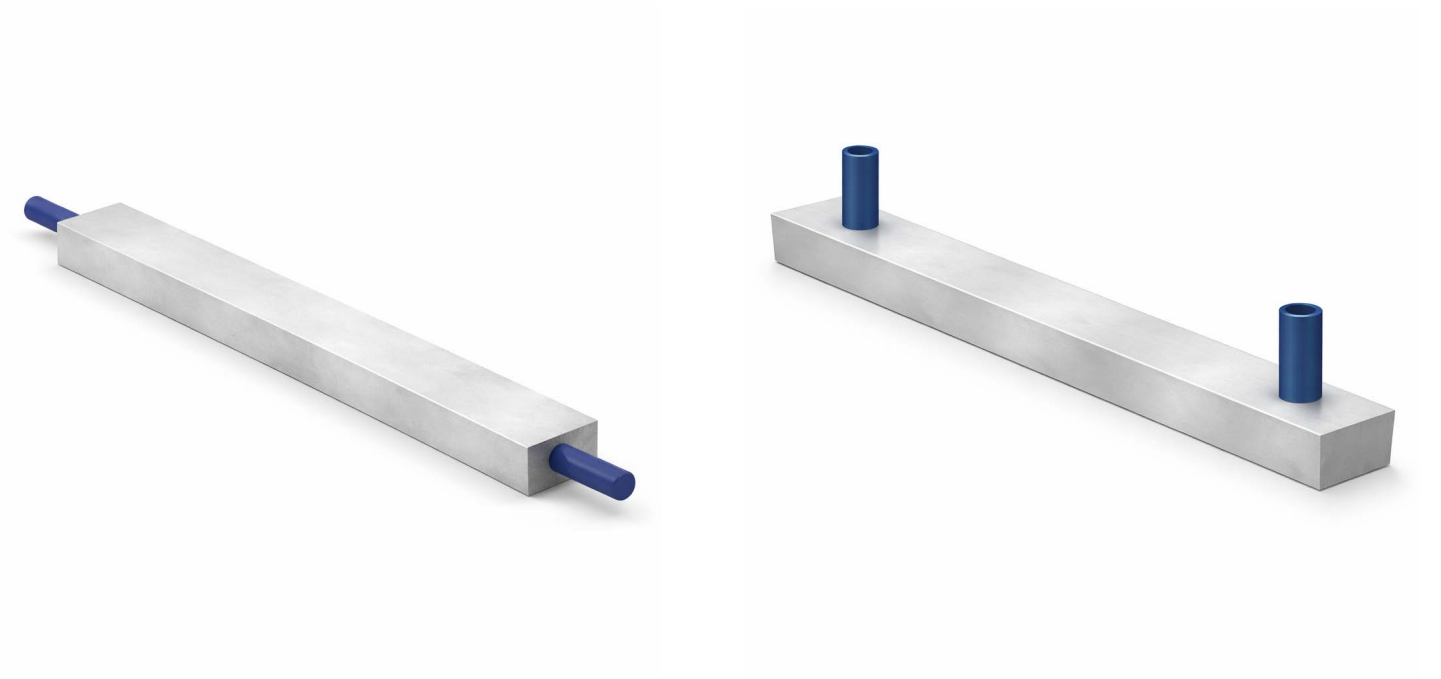


Platform Straight Core — Type A & Type B



NAME	LA (IN)	LC (IN)	W (IN)	H (IN)	NET WT (LBS)	GROSS WT (LBS)	CORE
PF_180_SBC	60	30	6½	5⅝	180	196	2" SCH.80
PF_285_SBC	120	90	5½	5⅝	285	340	2" SCH.80
PF_325 (0-3)	60-120	30-90	5½-8¼	5¾-7¾	325	374-411	3" SCH.80
PF_407 / 411	72-96	42-66	7-8	7-8½	407-411	480-484	3" SCH.80
PF_540 (0-1)	96-98	66-68	6½-8	7¾-9¾	540	612-613	3" SCH.80
PF_725 (0-3)	95-98	65-68	9¾	9⅝-9¾	725	878-881	4" SCH.80
PF_1050_SBC	144	114	9	9	1,050	1,187	4" SCH.80
PF_1100_SBC	122	92	10½	10	1,100	1,218	4" SCH.80
PF_1250_SBC	144	114	10	10½	1,250	1,387	4" SCH.80

Standard sizes shown. Any weight and size required by the customer can be manufactured. When ordering, please specify metal and chemical composition.



PRODUCT CATEGORY

Bracelet Anodes

Houston Anodes aluminum bracelet anodes are cast in low-iron, high-purity aluminum alloys, and can be manufactured in a wide variety of diameters, using our extensive inventory of water-cooled molds.

Configurations

- Square Weld-on
- Square Bolt-on
- Tapered Weld-on
- Tapered Bolt-on

Diameter Range

4½" – 48" (standard). Custom diameters available.

Weight Range

22 – 265 lbs (standard). Custom weights available.

ALUMINUM ALLOY

ZINC ALLOY

RETROFIT-READY



[Bracelet Anode Photo]



Bracelet Square Weld-on



Representative sizes from full catalog. Dimensions: Ø = pipe diameter, L = length, G = gap, T = thickness.

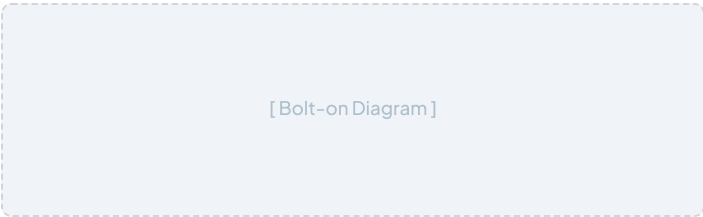
Ø (IN)	L (IN)	G (IN)	T (IN)	NET WT (LBS)	GROSS WT (LBS)	CORE
4½	10½-12	1¼-1½	1½	24-29	27-32	3/16 x 1
6¾	11¼-16½	1-1½	1¼-1½	37-43	46-52	3/16 x 1½
8-8¾	13-16	1-2	1½-2½	50-124	59-133	3/16 x 1½
10¾	10-20½	1½-3	1½-2½	73-152	88-167	¼ x 2
12¾	10½-20½	2-3	1½-2	80-190	98-208	¼ x 2
16-18	10-22	2-3½	1¼-2½	80-230	102-252	¼ x 2
20	13¾-22¾	2-3	1¼-3	166-260	186-280	¼ x 2
24	6¾-20	1½-3	1½-4	159-497	184-522	¼ x 2
30	12-15½	2½-3½	1½-2	164-284	192-312	¼ x 2
36	8¾-27	2-4	1¼-3	144-673	176-705	¼ x 2
42-48	7¾-18½	2-4	1½-3	322-442	365-482	¼ x 2

Standard sizes shown. Any weight and size required by the customer can be manufactured.

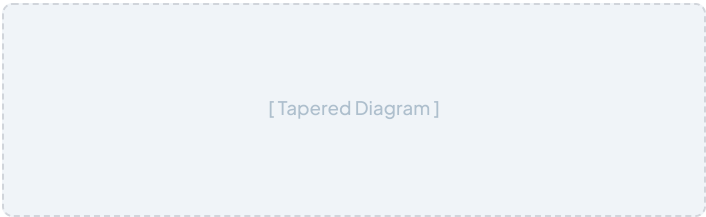


Bracelet Square Bolt-on & Tapered Configurations

Square Bolt-on



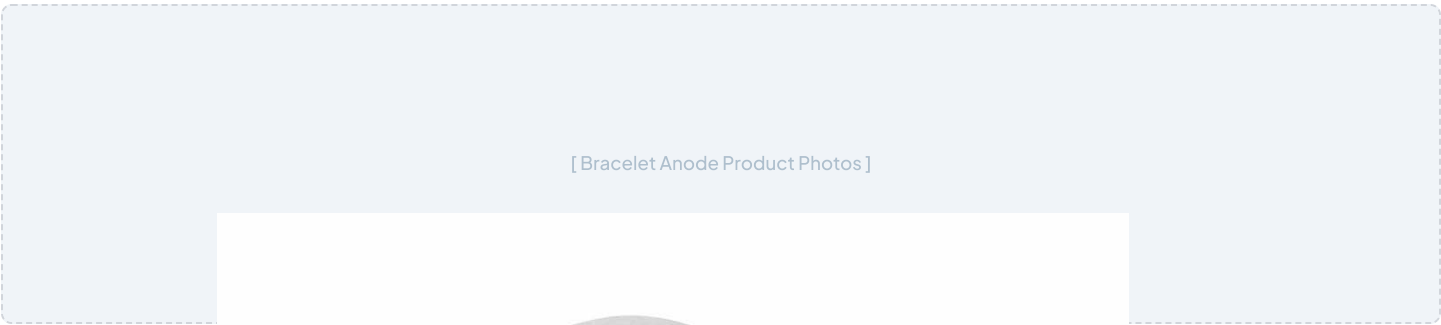
Tapered Weld-on / Bolt-on



Bolt-on bracelets feature drilled holes for bolt attachment. Tapered configurations provide optimized fit for specific pipeline geometries. Available in weld-on and bolt-on variants.

Available diameters: 4½" through 48". **Weight range:** 22–265 lbs standard. All configurations available in aluminum and zinc alloys.

Custom manufacturing: Houston Anodes can manufacture bracelet anodes in any diameter, length, gap width, and thickness to meet your exact pipeline specifications. Please contact us for advice on optimizing a bracelet anode to suit your specific requirements.



PRODUCT CATEGORY

Pier, Piling & Ballast

Houston Anodes aluminum galvanic anodes for piers and pilings are cast in one of two premium aluminum alloys, for optimal performance in any conditions. Core length comes standard depending on size, and holes may be drilled for bolt-on applications.

Configurations

- Trapezoidal Flush Core
- Trapezoidal Stand-off Core Type A
- Trapezoidal Stand-off Core Type B
- Trapezoidal Rod Core
- Trapezoidal Rod Stand-off Core

ALUMINUM ALLOY

ZINC ALLOY

BOLT-ON AVAILABLE



[Pier / Piling Anode Photo]



Trapezoidal Flush Core & Stand-off Core

[Flush Core Diagram]

[Stand-off Core Diagram]

Flush core anodes mount directly against the piling surface. Stand-off cores (Type A & B) provide clearance between the anode and the structure for enhanced current distribution. Available in multiple standard weights and core configurations.

Houston Anodes can cast anodes for piers, jetties, or any other piling and ballast to the exact specifications required. Please contact us for advice on optimizing an anode to suit your specific requirements.

[Pier/Piling Anode Product Photos]



Trapezoidal Rod Core & Rod Stand-off Core

[Rod Core Diagram]

[Rod Stand-off Core Diagram]

Rod core configurations utilize steel rod inserts for structural integrity. Rod stand-off cores combine rod mounting with standoff distance for optimal cathodic protection current distribution on pilings and ballast structures.

Note: Weight based on aluminum alloy; also available in zinc alloy. When ordering, please specify metal and chemical composition. Standard sizes shown — any weight and size required by the customer can be manufactured.

[Rod Core Anode Product Photos]

PRODUCT CATEGORY

Tanks & Vessels

Houston Anodes manufactures aluminum and zinc sacrificial anodes engineered specifically for internal vessel applications — protecting against corrosion in the harshest conditions. Vessel and tank anodes are cast in one of two premium aluminum alloys.

Product Types

- Heater Treater
- Heater Treater Eyebolt
- Bowling Ball
- Drum

Heater Treater Anodes can be supplied with Red/Blue caps and nuts upon customer's request. Houston Anodes can cast anodes for any sort of tank bottom or vessel interior to exact specifications.

ALUMINUM

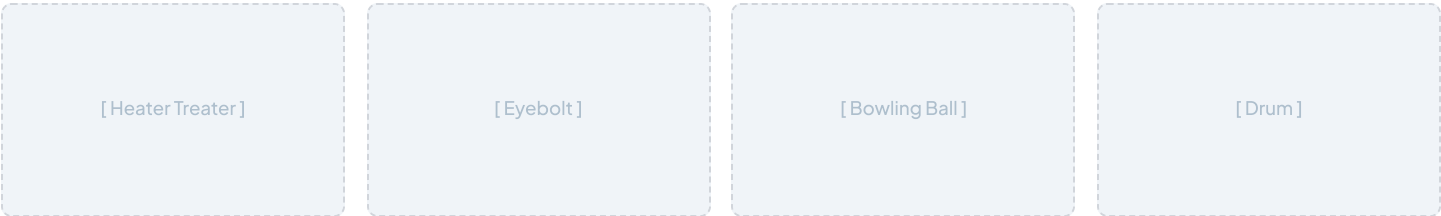
ZINC

CUSTOM INTERNALS

RED/BLUE CAPS

[Heater Treater / Tank Anode Photo]

Heater Treater, Eyebolt, Bowling Ball & Drum



Heater Treater

Standard and eyebolt configurations for internal vessel protection. Available with Red/Blue caps and nuts. Core length standard depending on size.

Bowling Ball

Spherical anodes designed for tank bottom protection. Cast in premium aluminum alloy for maximum cathodic protection output.

Heater Treater Eyebolt

Eyebolt mounting for easy installation and replacement in heater treater vessels. Secure mounting in high-temperature environments.

Drum

Cylindrical anodes for drum and vessel internal applications. Custom sizing available for any vessel configuration.

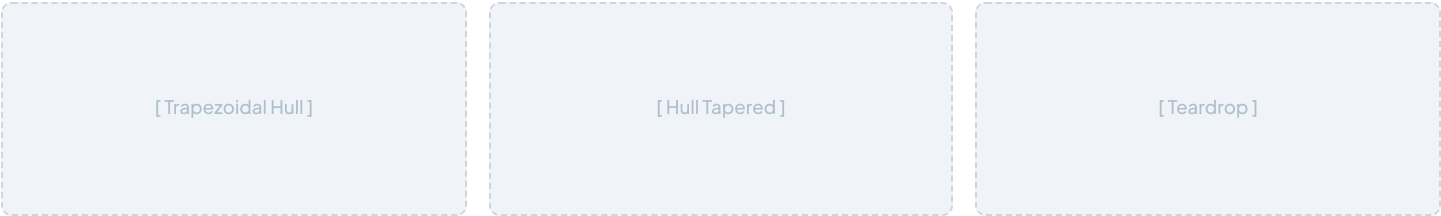
Custom vessel protection: Houston Anodes can cast anodes for any sort of tank bottom or vessel interior to the exact specifications required. Contact our engineering team for custom anode optimization tailored to your specific vessel requirements.



PRODUCT CATEGORY

Hull Anodes

Zinc hull anodes for marine equipment and vessel protection. Available in multiple configurations for new construction and retrofit applications. U.S. Navy anodes with proven 35-year service life.



Trapezoidal Hull

Standard and cast-in core configurations for hull bottom protection.

Trapezoidal Hull Cast-in Core

Integral core cast directly into the anode body for enhanced structural integrity.

Hull Tapered

Tapered profile for hydrodynamic efficiency on hull surfaces.

Tapered Trapezoidal

Combined taper and trapezoidal cross-section for optimized protection.

Teardrop

Hydrodynamic teardrop shape minimizes drag while providing effective cathodic protection.

U.S. Navy
35-Year Proven
Service Life



CUSTOM ENGINEERING

Custom Anodes & Offshore Wind

Custom Anode Types

Houston Anodes can fabricate any type, any size and any shape of anode to accommodate customer-specific needs. We also manufacture our own molds for complete production control.

[Clamp]

Clamp

[Mat]

Mat

[Link]

Link

Any geometry manufactured to exact specifications. Available in aluminum and zinc alloys. Custom weights from 5 lbs to custom configurations.

Offshore Wind Turbine Anodes

Houston Anodes has manufactured sacrificial anodes for major U.S. offshore wind farm projects:

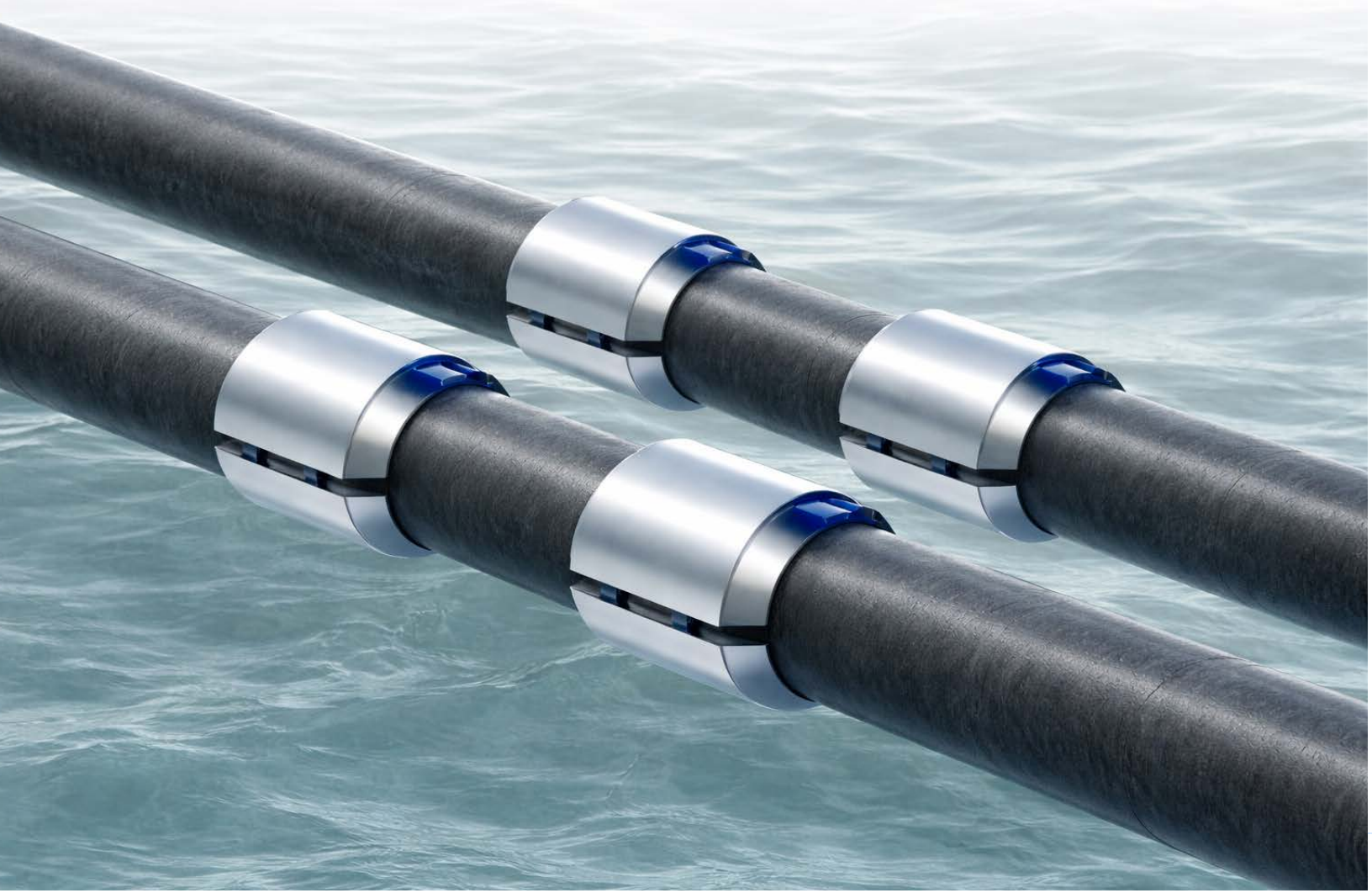
Ocean Wind 1 — New Jersey

Turbine foundation protection anodes

Revolution Wind — Rhode Island

Turbine foundation protection anodes

[Offshore Wind Turbine Anode Photo]



HOUSTON ANODES GUARANTEE

Certifications & Standards

Homogeneous alloys always conforming to exact international quality standards and specifications. Not a single product claim in 47 years.

STANDARD	DESCRIPTION
ISO 9001:2015 (DNV)	Quality management system for manufacturing of sacrificial anodes
ISO 17025	Laboratory accreditation — in-house testing laboratory
NACE SP-0387 / SP-0492	Metallurgical/inspection standards for sacrificial anodes
DNV RP B401	Cathodic protection design for offshore structures
ISO 15589-2:2012	Petroleum and natural gas industries — cathodic protection
NORSOK M-503	Norwegian petroleum industry cathodic protection standard
NACE TM-0190	Electrochemical testing protocol — performed on every batch

Cathodic Protection Services

- CP-1 CATHODIC PROTECTION
- CP-2 CATHODIC PROTECTION
- CP-4 CP SPECIALIST
- NACE CIP-1 COATING INSPECTOR
- NACE CIP-2 COATING INSPECTOR
- CP DESIGN & ENGINEERING

[Certifications / Lab Accreditation Certificates Photo]

GLOBAL TRUST, PROVEN RESULTS

Markets & Success Stories

Oil & Gas

Deepwater platforms, subsea structures, pipelines, refineries, chemical plants, FPSO/FSO

Offshore Wind

Turbine foundations, monopiles. Ocean Wind 1 (NJ) & Revolution Wind (RI)

Infrastructure

Piers, pilings, harbor facilities, ballast systems, marine structures

Industrials

Heater treaters, tanks, vessels, drums, corrosion protection

Defense (U.S. Navy)

Hull anodes. 35-year proven service life. Active defense approvals.

Custom & Special

Wind turbines, clamp, mat, link, any geometry to specification

Trusted by Industry Leaders

Houston Anodes currently protects major offshore assets and subsea structures throughout the world for valued clients including among others:

ExxonMobil, Shell, BP, Chevron, PEMEX, TechnipFMC, Subsea 7, J. Ray McDermott, Hess, Marathon, Williams, Anadarko, Oceaneering, Bredero Shaw, FloaTEC, Oil States Industries, Sapura Energy, Ørsted, Deepwater, Noble, ATP Oil & Gas, LLOG Exploration, ICA Fluor, Dragados Offshore, Global Industries, Ljungström, ARVOS Group, Corrpro (Aegion), Swiber, United States Navy

80%

PEMEX anodes - 35+ yrs

35 yr

U.S. Navy service life

1,000+

Projects in 5 years

FMC

Approved Global Supplier

GET IN TOUCH

Houston Anodes

Protecting the Infrastructure That Powers the World

47+

YEARS

1,000+

PROJECTS

0

CLAIMS

35 yr

NAVY SERVICE LIFE

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ISO 9001:2015

ISO 17025

DNV

NACE

100% USA